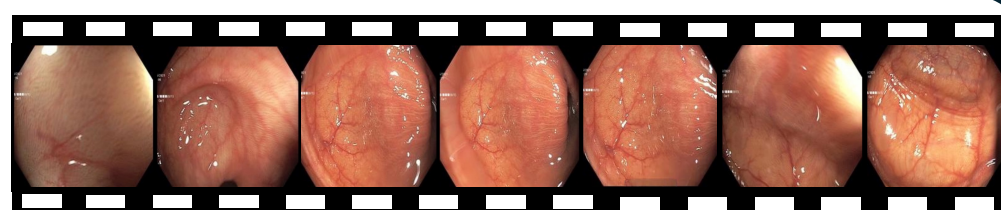


Query on Long Surgical Video



Q1: Is blue dye present on the mucosa?

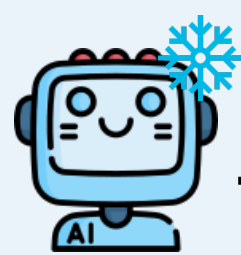
Q2: Is clean mucosa visible in this sequence?
.....

Temporally-Grounded Answers



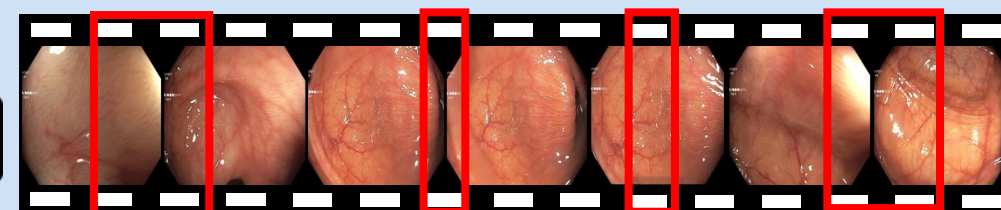
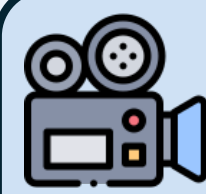
A1: Yes, the blue dye is on the mucosa.

A2: No, the clean mucosa visible in this sequence.
.....

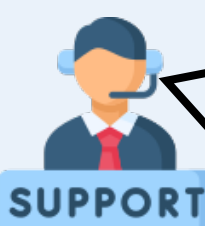


Search

VLM



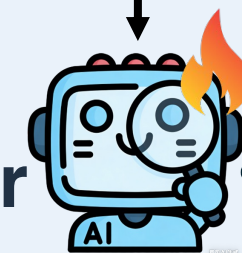
[[0.4, 0.56, 0.6], [3.20, 3.84, 4.16], [5.76, 6.04, 6.20], [8.46, 8.66, 9.10], ...]



SUPPORT

Please estimate the possible happening time intervals for questions in the video.

Policy Instructor

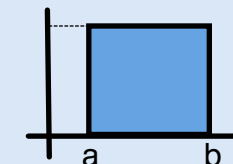


How to Re-sample?

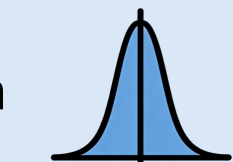


Sampling Policies

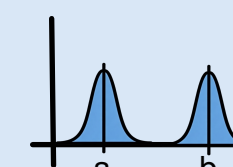
Uniform



Gaussian

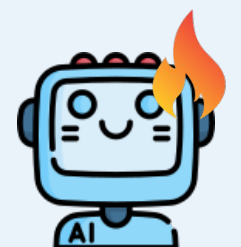


U-Shape

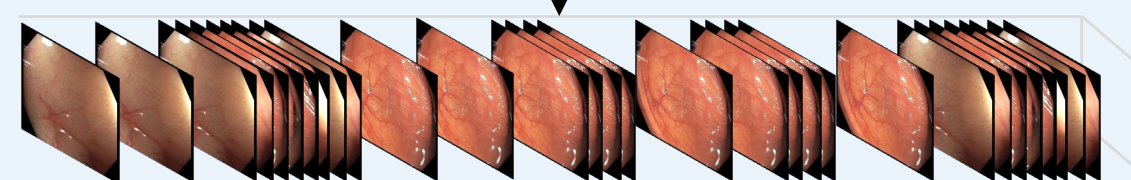


.....

Inferring



VLM



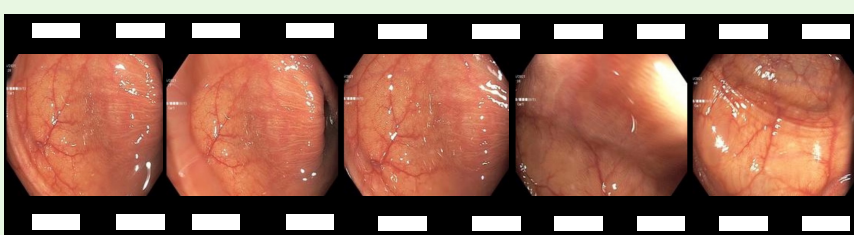
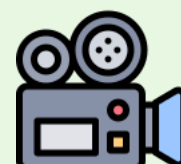
Re-sampled Sequence

Temporally-Grounded Multi-Policy Scaling

Timestamp Tokenizer



0.1s 0.3s 0.5s 0.7s 0.9s



Input Sequences

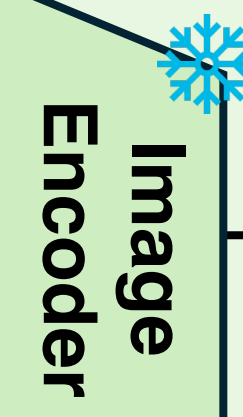


Image Encoder

Masked Tokens

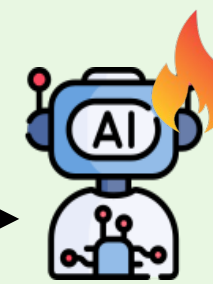


Token Compression

Compact Representation



Retriever



LLM (w/ LoRA)

Faithful Temporal Consolidation

$\mathcal{L}_{ret}$